12 Securing IoT communications

This chapter covers

- Securing IoT communications with Datagram TLS
- Choosing appropriate cryptographic algorithms for constrained devices
- Implementing end-to-end security for IoT APIs
- Distributing and managing device keys

So far, all the APIs you've looked at have been running on servers in the safe confines of a datacenter or server room. It's easy to take the physical security of the API hardware for granted, because the datacenter is a secure environment with restricted access and decent locks on the doors. Often only specially vetted staff are allowed into the server room to get close to the hardware. Traditionally, even the clients of an API could often be assumed to be reasonably secure, because they were desktop PCs installed in an office environment. This has rapidly changed as first laptops and then smartphones have moved API clients out of the office environment. The *internet of things* (IoT) widens the range of environments even further, especially in industrial or agricultural settings where devices may be deployed in remote environments with little physical protection or monitoring. These IoT devices talk to APIs in messaging